OPT benchmarks: resource and latency impact

Updated 2026-10-13T13:22:21+02:00. Host: hacc-u55c-02; U55C, PCIe Gen3 x16; 250 MHz. All board runs completed.

Eight independently built OPT variants use the same model and weights. OPT_1_STREAMING is the comparison baseline. R_S is excluded from these measurements. Nonstream buffers all 65536 pixels and waits for raw-input completion before CNN consumption.

Label	Iteration	Board state
Nonstream	OPT_0_NONSTREAMING	passed
Streaming	OPT_1_STREAMING	passed
A 100k	OPT_3_BRAM_FIFO_A	passed
FIFO 1024	OPT_FIFO_A_P1024	passed
FIFO 16384	OPT_FIFO_A_P16384	passed
B2	OPT_2_DSP_RESOURCES_B2	passed
B2+72	OPT_4_DSP_B2_CONV34_RF72	passed
B2+72/288	OPT_5_DSP_B2_RF72_RF288	passed

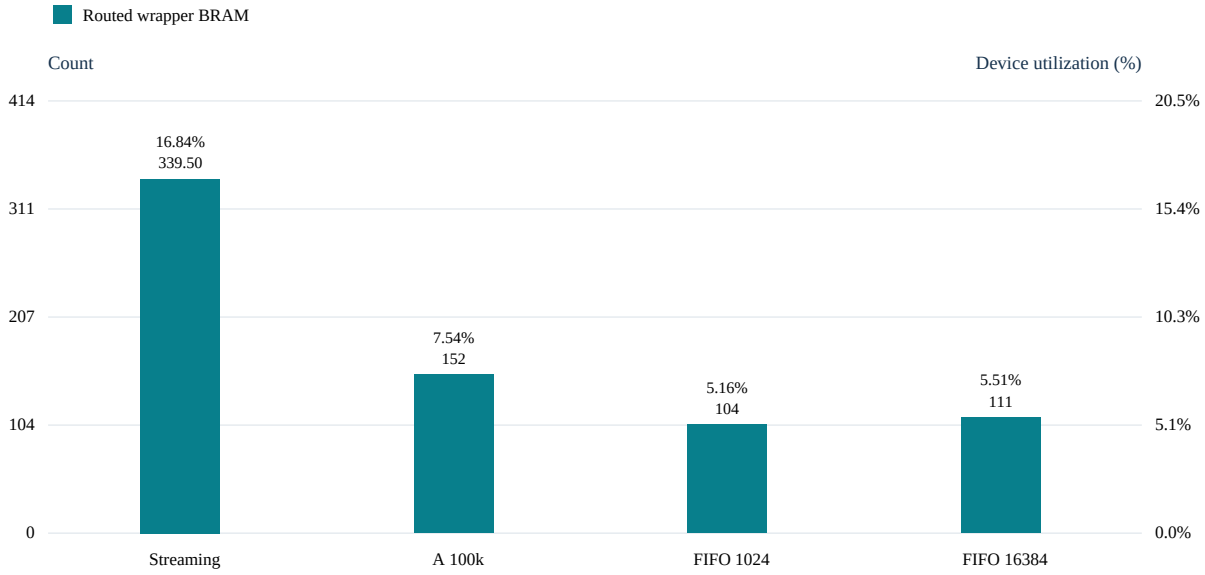
Protocol: exact model logits and verdicts; 520 admission calls across three sessions per image; pilot before timing; 20 warm-ups + 200 measured trials per workload and scope. 15 timed workloads: eight real bitstreams, four synthetic sizes and three preprocessing boundary cases. The streaming baseline is repeated at the end.

Scope: file e2e starts before file open/read and ends when the host observes the verdict. It includes input copies, submission, transfer, FPGA preprocessing and CNN work. It excludes initial allocation/programming and diagnostic CSR readback. Filesystem cache is warm; this does not measure cold-storage latency. Resident-input latency excludes file reading and preparation.

A / FIFO variants: BRAM reduction

Variant	BRAM tiles	Device %	Reduction vs baseline
Streaming	339.5	16.84%	0.00%
A 100k	152	7.54%	55.23%
FIFO 1024	104	5.16%	69.37%
FIFO 16384	111	5.51%	67.30%

BRAM: native FIFO optimization



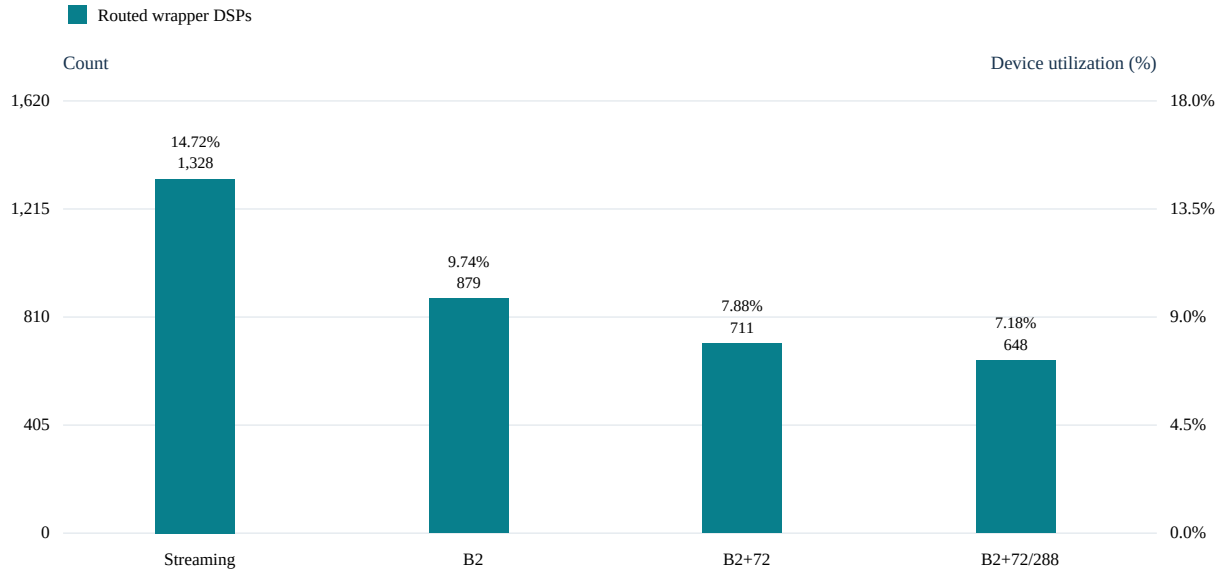
Left axis: tiles; right axis and labels: device utilization.

Profiling depths are temporary native profiling capacities; final internal FIFO depths differ by channel. Profiles were reused, not rerun. A 100k, FIFO1024 and FIFO16384 use zero URAM; the streaming baseline uses 11 URAM blocks.

B2 variants: DSP reduction

Variant	conv3/4/5/6 reuse	DSPs / device %	Reduction vs baseline
Streaming	36/36/32/32	1328 / 14.72%	0.00%
B2	36/36/144/144	879 / 9.74%	33.81%
B2+72	72/72/144/144	711 / 7.88%	46.46%
B2+72/288	72/72/288/288	648 / 7.18%	51.20%

DSP: increasing multiplier reuse



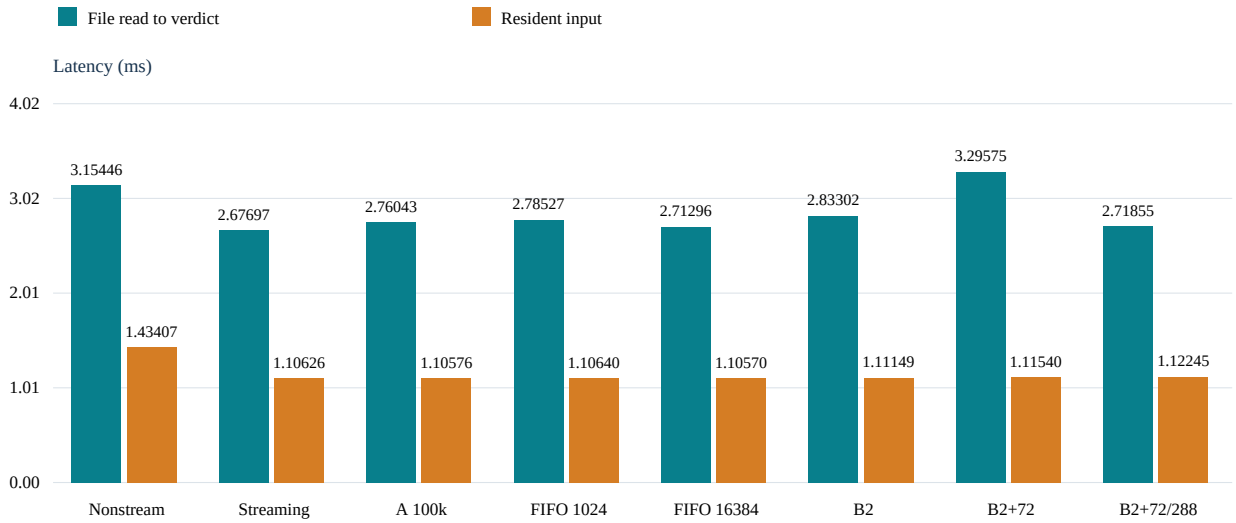
Left axis: DSP count; right axis and labels: device utilization.

B variants preserve baseline FIFOs. No A+B combination is included yet. Select the combination after comparing measured latency and both resource costs.

End-to-end latency across all variants

Variant	File e2e median / p95 (ms)	Resident median / p95 (ms)
Nonstream	3.1545 / 3.3718	1.4341 / 1.4351
Streaming	2.6770 / 2.8833	1.1063 / 1.1072
A 100k	2.7604 / 2.9657	1.1058 / 1.1070
FIFO 1024	2.7853 / 2.9374	1.1064 / 1.1070
FIFO 16384	2.7130 / 2.9821	1.1057 / 1.1069
B2	2.8330 / 3.0349	1.1115 / 1.1124
B2+72	3.2958 / 4.5826	1.1154 / 1.1165
B2+72/288	2.7186 / 2.8998	1.1225 / 1.1234

4 MiB: measured host latency



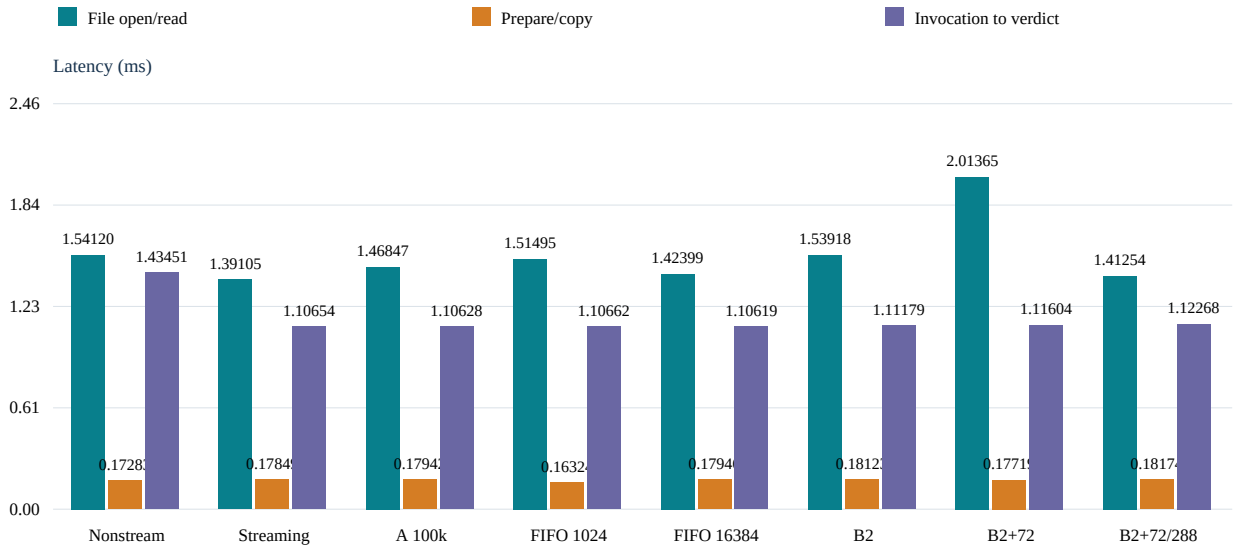
Median milliseconds. Each scope has its own measured trials.

Both scopes transmit the entire bitstream for every call. Resident-input results isolate the hardware path more closely; file e2e includes filesystem and copy costs. The same raw input bytes and preparation semantics are used in every variant.

File e2e: reading, preparation and inference

Variant	File read	Input preparation	Invocation to verdict	Complete file e2e
Nonstream	1.5412	0.1728	1.4345	3.1545
Streaming	1.3910	0.1785	1.1065	2.6770
A 100k	1.4685	0.1794	1.1063	2.7604
FIFO 1024	1.5150	0.1632	1.1066	2.7853
FIFO 16384	1.4240	0.1795	1.1062	2.7130
B2	1.5392	0.1812	1.1118	2.8330
B2+72	2.0137	0.1772	1.1160	3.2958
B2+72/288	1.4125	0.1817	1.1227	2.7186

4 MiB: where file-to-verdict time is spent



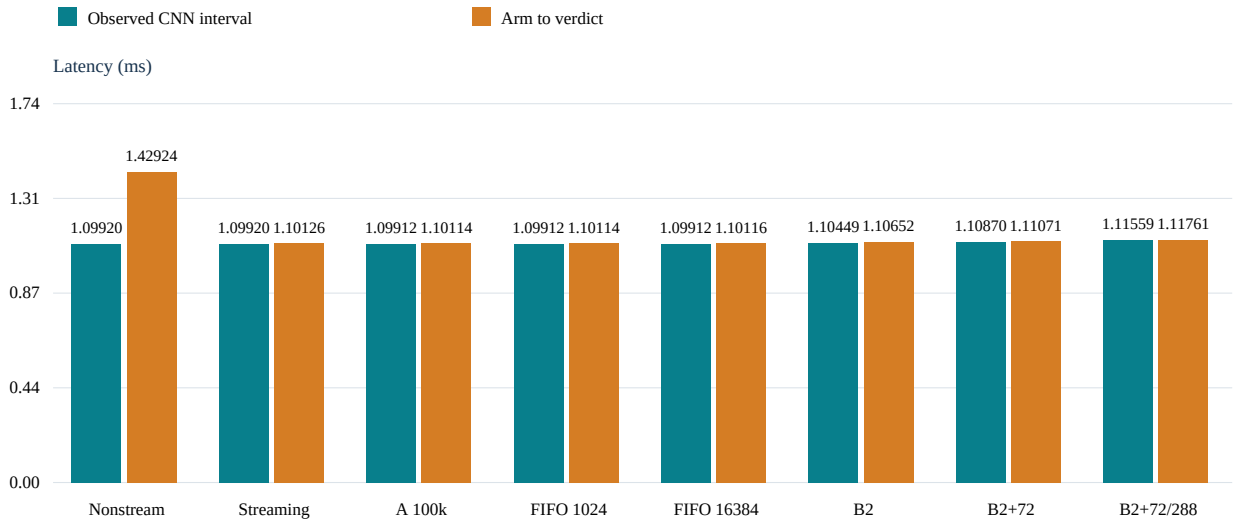
Median ms from the file-scope trials. Medians are not strictly additive.

File read includes allocation of the input array and filesystem open/read/close. Input preparation includes framing, copying to the registered buffer and setting raw length. Invocation includes transfer, FPGA preprocessing/CNN and verdict observation. Small gaps between these operations are included in complete file e2e. The file read and copy phases occur before submission; FPGA preprocessing and CNN may then overlap.

FPGA latency and streaming overlap

Variant	Arm to verdict	CNN first input to score	Preprocessing	Overlap
Nonstream	1.4292	1.0992	0.3284	0.0000
Streaming	1.1013	1.0992	0.3295	0.3291
A 100k	1.1011	1.0991	0.3295	0.3291
FIFO 1024	1.1011	1.0991	1.0587	1.0583
FIFO 16384	1.1012	1.0991	0.8125	0.8121
B2	1.1065	1.1045	0.3295	0.3291
B2+72	1.1107	1.1087	0.3295	0.3291
B2+72/288	1.1176	1.1156	0.3295	0.3291

4 MiB: FPGA timestamp intervals

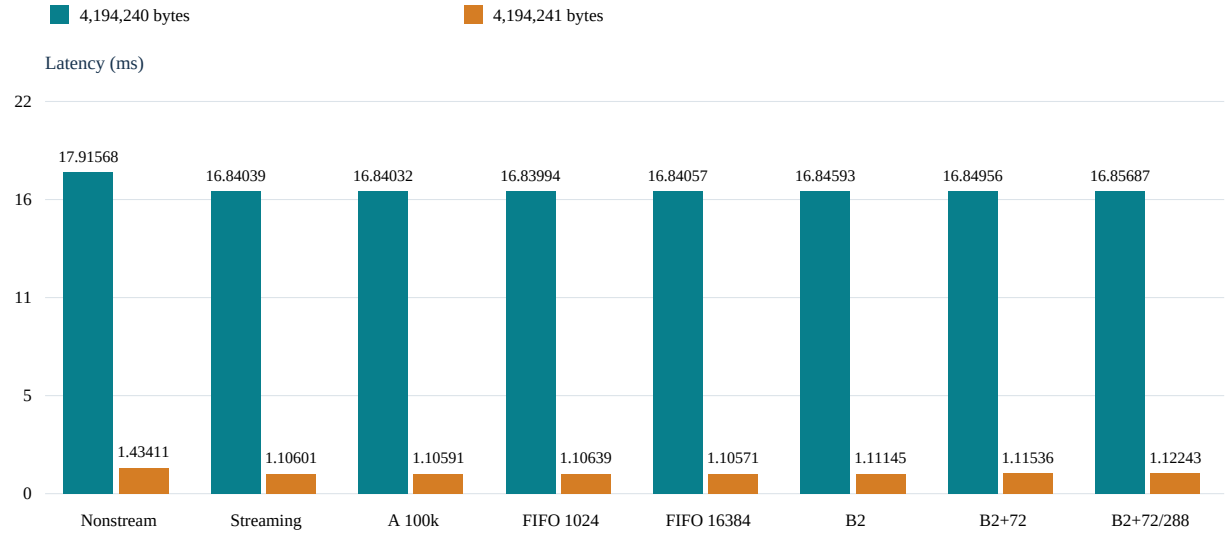


Median ms, calibrated from 32 bracketed clock samples before/after each block.

CNN first-input-to-score is the observed interval, including pixel starvation/backpressure. It is not standalone CNN compute time. Streaming reception, preprocessing and CNN intervals overlap and must not be summed. Nonstream has no preprocessing/CNN overlap; its full frame fill appears in arm-to-verdict.

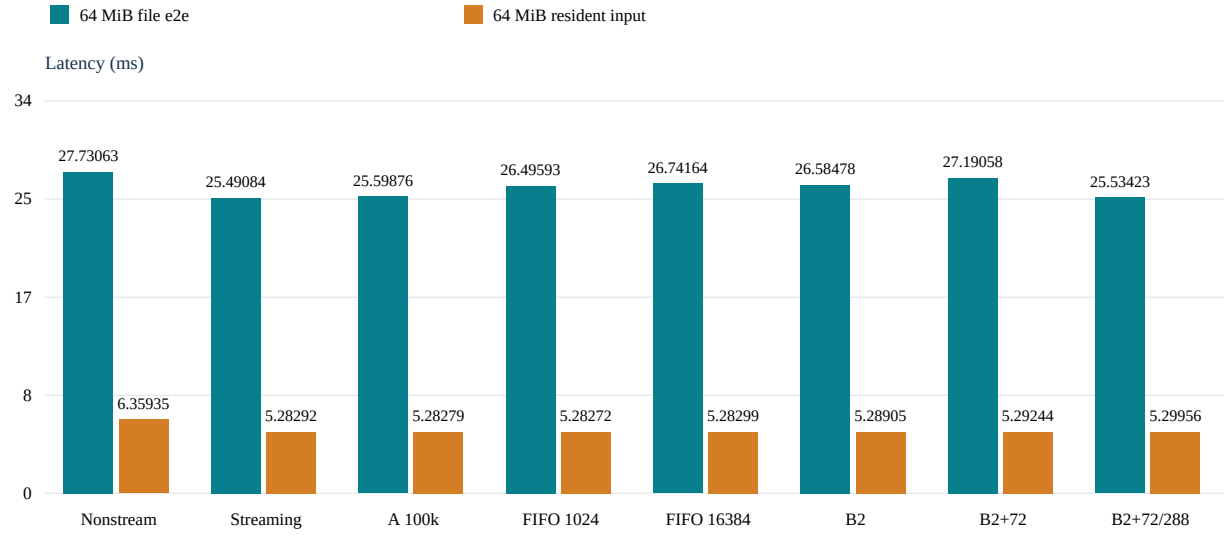
Preprocessing boundary and input size

One-byte boundary: observed host latency



Median ms, prepared input. The preprocessing implementation changes path here.

64 MiB: file reading and copies matter



Median ms. Both modes send the complete raw bitstream.

A size-dependent latency discontinuity can arise in preprocessing even with identical CNN weights. File e2e also includes host cache and copy variability; small differences between variants should be interpreted alongside resident-input and FPGA intervals, and the final baseline repeat.

File-to-verdict latency: all workloads

Median milliseconds; 200 measured trials/case. Boundary cases 4,194,240 / 4,194,241 / 4,194,242 bytes exercise the preprocessing branch transition. Raw CSV and JSON retain p95 and per-trial results.

Input bytes	Nonstream	Streaming	A 100k	FIFO 1024	FIFO 16384	B2	B2+72	B2+72/288
4,194,240	19.4029	18.3963	18.4254	18.4154	18.4361	18.4212	18.8406	18.4372
4,194,241	3.1214	2.7953	2.7228	2.7069	2.7115	2.7311	3.2958	2.8596
4,194,242	2.9963	2.8122	2.7399	2.7869	2.7878	2.7917	3.3687	2.8124
4,194,304	3.1545	2.6770	2.7604	2.7853	2.7130	2.8330	3.2958	2.7186
5,540,996	3.4480	2.9516	3.0189	2.9941	2.9814	3.0451	3.5969	3.0984
12,004,440	6.0510	5.1772	5.1499	5.1909	5.1529	5.1741	5.5543	5.1125
16,777,216	7.0004	5.7870	5.6691	5.7621	5.7454	5.9356	6.6122	5.9156
18,876,484	7.8613	6.7462	6.6144	6.8325	6.6633	6.8801	7.5790	6.8176
30,323,588	11.6862	10.7583	10.8848	10.8854	10.8551	10.9567	11.3053	10.6400
34,162,708	15.7805	14.6299	14.6748	14.5547	14.6629	14.6401	15.3538	14.5488
41,278,872	17.8838	16.7446	16.6458	16.7617	16.8147	16.6807	17.2433	16.8005
48,540,364	20.4466	19.7574	19.4989	19.7018	19.6395	19.5467	20.0817	19.4150
55,676,772	23.5091	21.4351	21.4692	22.4022	22.4511	22.4178	23.1213	21.4485
67,108,864	27.7306	25.4908	25.5988	26.4959	26.7416	26.5848	27.1906	25.5342
134,217,664	52.2431	51.1573	51.2274	51.0596	51.1392	51.3575	51.8040	51.1629

Resident-input latency: all workloads

Median milliseconds; 200 measured trials/case. Boundary cases 4,194,240 / 4,194,241 / 4,194,242 bytes exercise the preprocessing branch transition. Raw CSV and JSON retain p95 and per-trial results.

Input bytes	Nonstream	Streaming	A 100k	FIFO 1024	FIFO 16384	B2	B2+72	B2+72/288
4,194,240	17.9157	16.8404	16.8403	16.8399	16.8406	16.8459	16.8496	16.8569
4,194,241	1.4341	1.1060	1.1059	1.1064	1.1057	1.1115	1.1154	1.1224
4,194,242	1.4341	1.1061	1.1059	1.1064	1.1057	1.1116	1.1157	1.1226
4,194,304	1.4341	1.1063	1.1058	1.1064	1.1057	1.1115	1.1154	1.1225
5,540,996	1.5396	1.1058	1.1062	1.1049	1.1054	1.1112	1.1156	1.1223
12,004,440	2.0455	1.1065	1.1065	1.1063	1.1055	1.1117	1.1153	1.1220
16,777,216	2.4191	1.3437	1.3435	1.3435	1.3436	1.3500	1.3532	1.3604
18,876,484	2.5836	1.5079	1.5077	1.5078	1.5079	1.5142	1.5174	1.5245
30,323,588	3.4795	2.4036	2.4034	2.4040	2.4034	2.4097	2.4131	2.4202
34,162,708	3.7802	2.7041	2.7039	2.7040	2.7040	2.7102	2.7136	2.7206
41,278,872	4.3373	3.2611	3.2614	3.2617	3.2610	3.2674	3.2705	3.2776
48,540,364	4.9057	3.8294	3.8292	3.8293	3.8294	3.8359	3.8390	3.8460
55,676,772	5.4644	4.3882	4.3884	4.3887	4.3879	4.3944	4.3977	4.4047
67,108,864	6.3593	5.2829	5.2828	5.2827	5.2830	5.2891	5.2924	5.2996
134,217,664	11.6118	10.5369	10.5361	10.5366	10.5359	10.5423	10.5454	10.5526

Baseline repeat and admission evidence

Input bytes	First baseline file e2e	Final baseline file e2e	Change
4,194,304	2.6770	2.7326	+2.08%
16,777,216	5.7870	5.7547	-0.56%
67,108,864	25.4908	25.5931	+0.40%
134,217,664	51.1573	51.0426	-0.22%

4 MiB interval	First baseline (ms)	Final baseline (ms)	Change
Resident host	1.1063	1.1059	-0.03%
FPGA arm to verdict	1.1013	1.1012	-0.00%
CNN input to score	1.0992	1.0992	+0.00%

Variant	Exact admission calls	Full measured calls	Image SHA256 prefix
Nonstream	520	6000	4ac162c13f4dfb8f
Streaming	520	6000	46ff53e3554f9c4e
A 100k	520	6000	fde068c8d73d0712
FIFO 1024	520	6000	a91972ffde24de50
FIFO 16384	520	6000	a8021484ecd4b6c5
B2	520	6000	8a84b715b22d4834
B2+72	520	6000	7a772fcca10fc2d7
B2+72/288	520	6000	6d66112d06e97346
Streaming repeat	520	6000	46ff53e3554f9c4e

A failed parity/event/clock check prevents admission of that variant. Successful results are tied to exact bitstream, source manifest, driver and software hashes. Standard driver reset is retained. CPU2 / NUMA0 and HPF allocation are used throughout. Any measured baseline drift is reported, not silently normalized away.

Appendix: all routed resource types

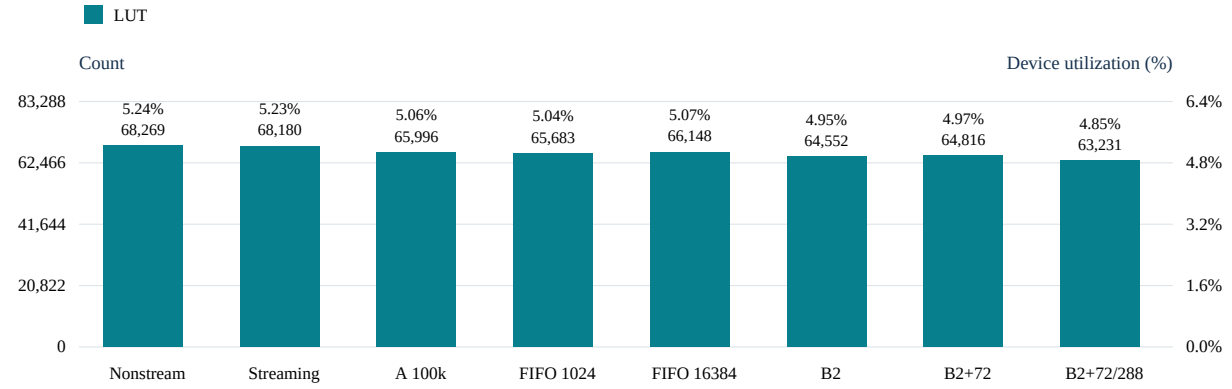
Variant	LUT	FF	BRAM_tiles	DSP	URAM
Nonstream	68269 / 5.24%	87137 / 3.34%	369.5 / 18.33%	1328 / 14.72%	11 / 1.15%
Streaming	68180 / 5.23%	87042 / 3.34%	339.5 / 16.84%	1328 / 14.72%	11 / 1.15%
A 100k	65996 / 5.06%	93477 / 3.59%	152 / 7.54%	1328 / 14.72%	0 / 0.00%
FIFO 1024	65683 / 5.04%	91729 / 3.52%	104 / 5.16%	1328 / 14.72%	0 / 0.00%
FIFO 16384	66148 / 5.07%	92641 / 3.55%	111 / 5.51%	1328 / 14.72%	0 / 0.00%
B2	64552 / 4.95%	84545 / 3.24%	322.5 / 16.00%	879 / 9.74%	11 / 1.15%
B2+72	64816 / 4.97%	82820 / 3.18%	309 / 15.33%	711 / 7.88%	11 / 1.15%
B2+72/288	63231 / 4.85%	81645 / 3.13%	290 / 14.38%	648 / 7.18%	11 / 1.15%

Each cell is count / device utilization. Wrapper scope includes integration and debug logic, excluding the static Coyote shell. BRAM tiles = RAMB36 + RAMB18 / 2. The nonstream full-frame buffer adds 30 BRAM tiles relative to the matching streaming baseline.

Full design	LUT	FF	BRAM_tiles	DSP	URAM
Nonstream	189158 / 14.51%	259187 / 9.94%	511.5 / 25.37%	1328 / 14.72%	11 / 1.15%
Streaming	188993 / 14.50%	258992 / 9.93%	481.5 / 23.88%	1328 / 14.72%	11 / 1.15%
A 100k	186868 / 14.33%	265459 / 10.18%	294 / 14.58%	1328 / 14.72%	0 / 0.00%
FIFO 1024	186350 / 14.29%	263665 / 10.11%	246 / 12.20%	1328 / 14.72%	0 / 0.00%
FIFO 16384	187000 / 14.34%	264608 / 10.15%	253 / 12.55%	1328 / 14.72%	0 / 0.00%
B2	185282 / 14.21%	256481 / 9.84%	464.5 / 23.04%	879 / 9.74%	11 / 1.15%
B2+72	185566 / 14.23%	254740 / 9.77%	451 / 22.37%	711 / 7.88%	11 / 1.15%
B2+72/288	184020 / 14.12%	253613 / 9.73%	432 / 21.43%	648 / 7.18%	11 / 1.15%

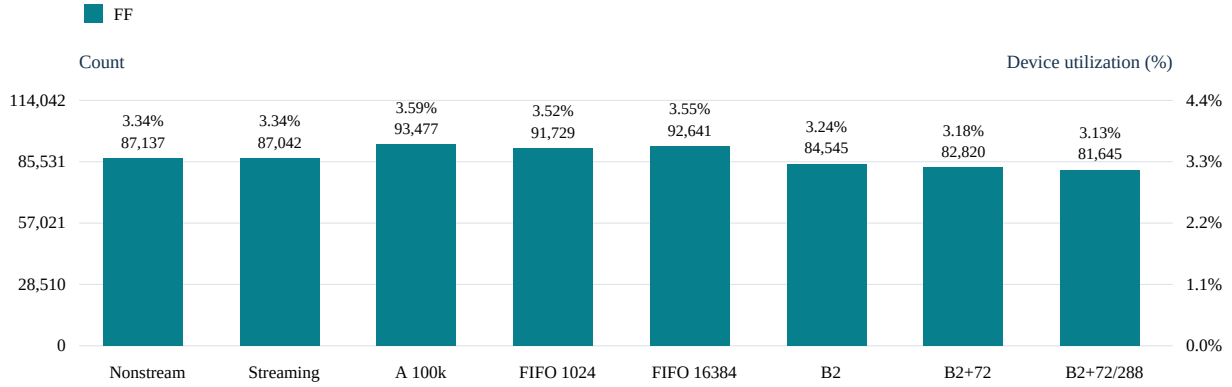
Appendix: resource plots

LUT across all variants



Left: count. Right: device utilization %. Bar labels show both.

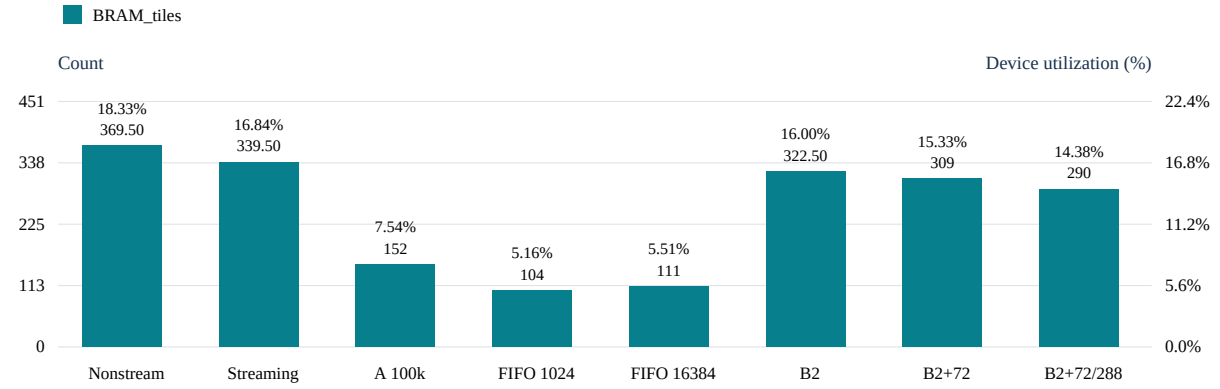
FF across all variants



Left: count. Right: device utilization %. Bar labels show both.

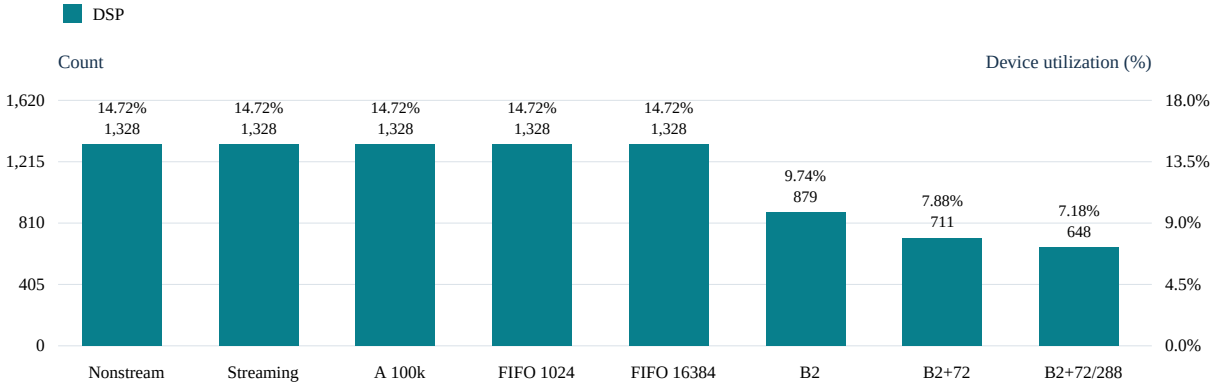
Appendix: resource plots

BRAM_tiles across all variants



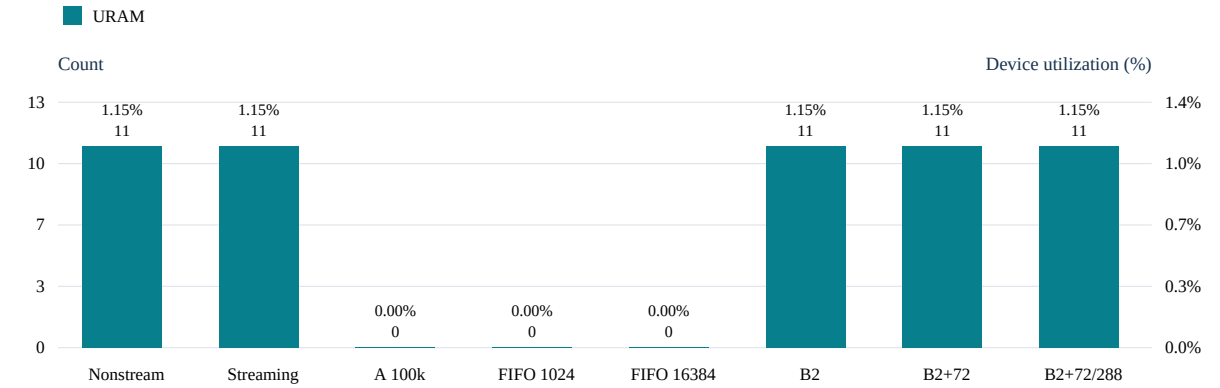
Left: count. Right: device utilization %. Bar labels show both.

DSP across all variants



Left: count. Right: device utilization %. Bar labels show both.

URAM across all variants



Left: count. Right: device utilization %. Bar labels show both.

Evidence and paths are recorded in the companion JSON. Production and historical report packages are preserved. No combined A+B build or measurement has been selected automatically.